

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
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- Conclusion
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Executive Summary

Summary of Methodologies

1) **Data Collection**:

- Gathered SpaceX launch data via API and web scraped data from Wikipedia.

2) **Data Cleaning and Preparation**:

- Cleaned and Formatted the Data.
- Stored the data in Db2 database and performed SQL queries.
- Performed Exploratory Data Analysis (EDA).
- Conducted Feature Engineering.

3) **Interactive Visualization:**

- Mapped launch sites and success rates using Folium.
- Built an interactive dashboard with Plotly Dash.

4) **Model Building And Evaluation:**

- Created SVM, Decision Trees and K-Nearest Neighbors.
- Tuned Hyperparameters with GridSearchCV.
- Evaluated Models using test data.

Summary of Results

1) **Data Insights:**

- Identified factors that influenced the Falcon 9 landings.
- Visualized geographical patterns and their success rates.

2) **Model Performance:**

- `SVM and K-Nearest Neighbors: 83.3% Accuracy`
- `Decision Tree: 94.4% Accuracy`

3) **Key Findings:**

- Factors that impacted the landing success rate were "**Launch Site**" and "**Payload Mass**"
- The **Decision Tree Model** performs the best based on, the highest accuracy on the test data.

Introduction

Project Background And Context:

Within this capstone project we aim to predict, the successful landing of the Falcon 9 first stage through analyzing the SpaceX launch data via the API and the historical records we web scraped from Wikipedia. We choose SpaceX as they advertise the Falcon 9 rocket launches to be significantly cheaper than their competitors, due to their ability to reuse the first stage of the rocket. Hence, by creating a model that accurately predicts whether the first stage lands, we can determine the cost of a launch. This information is valuable to alternative companies who would like to bid against SpaceX for a rocket launch.

Key Objectives:

What factors influence the successful landing of the Falcon 9?

Can we accurately predict the landing outcome of the Falcon 9 first stage via machine learning models?

Which machine learning model performs the best in predicting the Falcon 9 landing success?

Section 1

Methodology



Methodology

Executive Summary

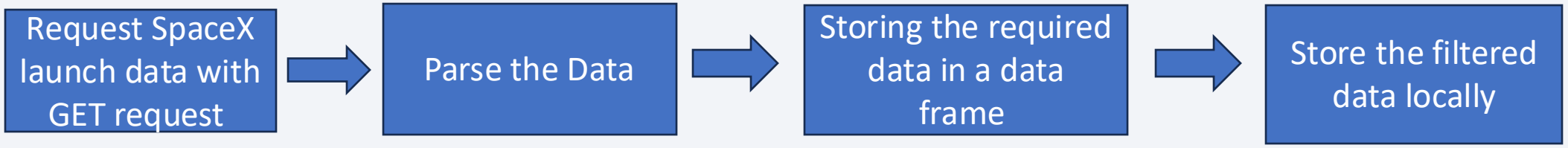
- **Data collection methodology:**
  - We gathered data from SpaceX API through GET requests then decoded the response content as a JSON, which was then converted into a Pandas dataframe.
  - We gathered data from Wikipedia via the BeautifulSoup which web scarped the HTML table and converted it into a Pandas dataframe.
- **Perform data wrangling:**
  - Data processing involved handling missing values, extracting key data points and applying one-hot encoding to categorical data.
- **Perform exploratory data analysis (EDA) using visualization and SQL:**
  - Used Matplotlib and Seaborn to visualize the relationship between factors such as launch success rates, payloads and launch sites.
  - Used SQL queries to gather key insights such as unique launch sites, payload mass and the total number of successful missions.
- **Perform interactive visual analytics using Folium and Plotly Dash:**
  - Used Folium to mark the successful/failed launches for each site on the map, then calculated the distances between the launch sites to its proximities.
  - Used Plotly Dash to build an interactive dashboard, where users can perform visual analytics on SpaceX launch data in real time via dropdown list and a range slider.

Methodology

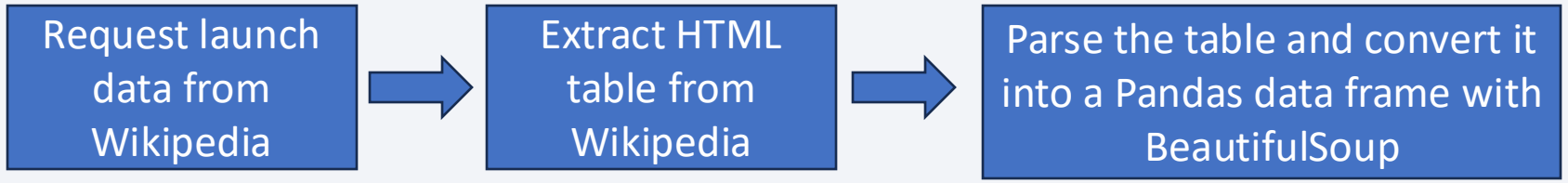
- **Perform Predictive Analysis Using Classification Models:**
  - Split the data set into training data and test data.
  - Built various classification models such as Logistic Regression, SVM, KNN and Decision Trees.
  - Evaluated the models based on accuracy to identify the best performing method.
  - Performed hyperparameter tuning using GridSearchCV to improve the model's accuracy.

Data Collection

Data Collection via SpaceX API Request:



Data Collection via Web Scraping Wikipedia:



Data Collection – SpaceX API

1. **Request SpaceX launch data:**

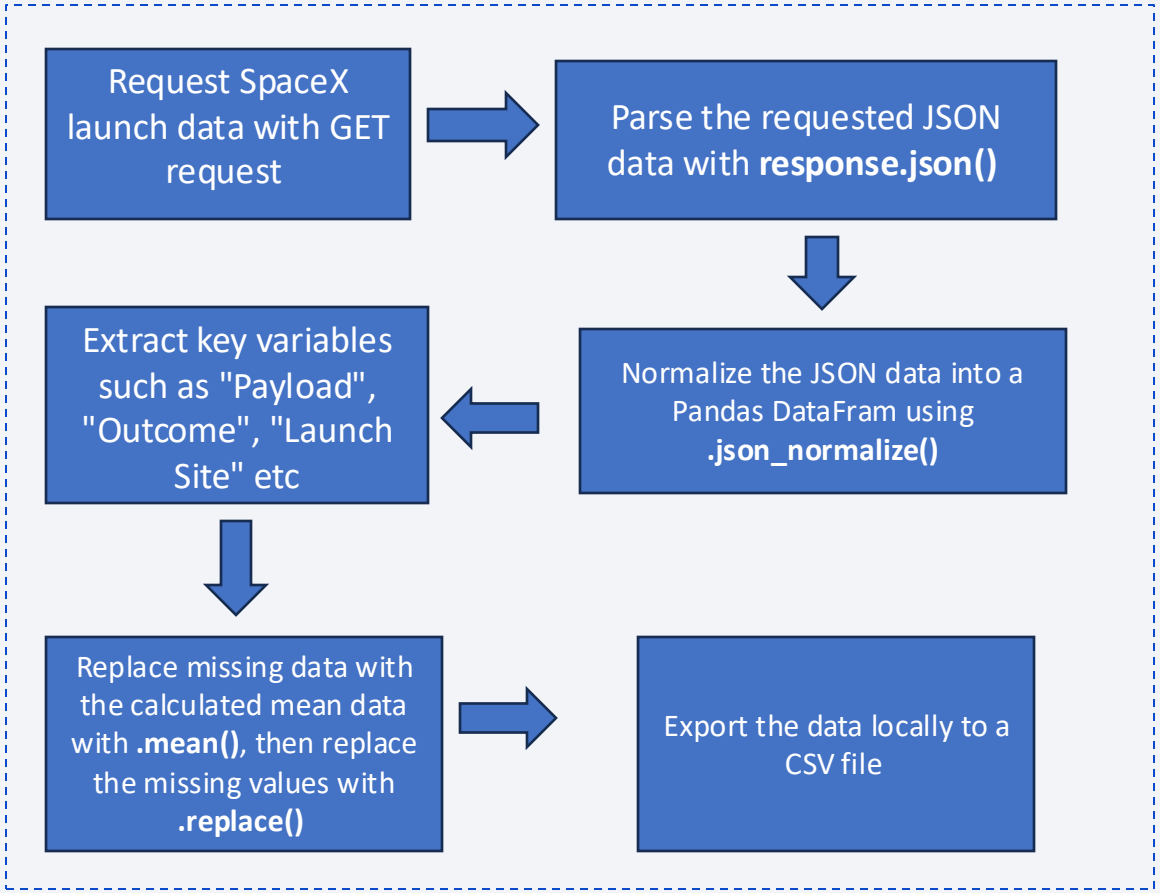
   - Request the rocket launch data from SpaceX API using the GET request.

2. **Parse the SpaceX launch data:**

   - Decode the API response as JSON using **.json()** and turn it into a Pandas DataFrame using **.json_normalize()**
   - Extract key variables such as "Payload Mass", "Launch Date", "Launch Site", "Outcome" and "Orbit".

3. **Sorting the data:**

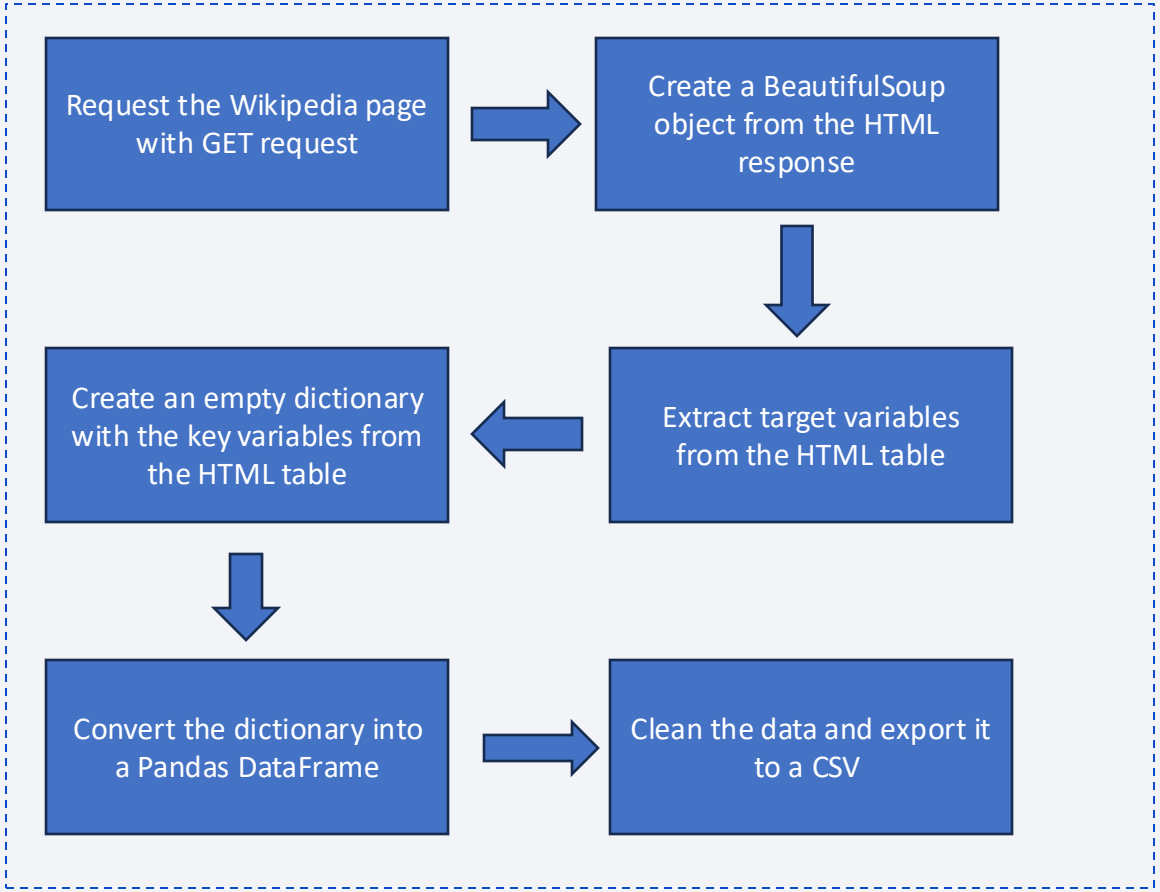
   - Store the filtered data locally for further analysis.



GitHub URL: [jupyter-labs-spacex-data-collection-api.ipynb]

Data Collection - Scraping

1. **Request launch data from Wikipedia:**
   - Perform an HTTP GET method to request the Falcon9 launch HTML page.
2. **Extract the HTML table:**
   - Create a BeautifulSoup object from the HTML response.
   - Extract target variables from the HTML table.
3. **Create a data frame from the HTML table:**
   - Create an empty dictionary with the key variables from the HTML table
   - Convert the dictionary into a Pandas DataFrame
   - Clean the data by replacing unexpected annotations and other types of noise
   - Export the DataFrame to a CSV and store it locally.



GitHub URL: [jupyter-labs-webscraping.ipynb]

Data Wrangling

During this project there were couple of missing data points within the data we acquired. In order to fix this, we performed Data Wrangling. Data Wrangling is a process that transforms raw unstructured data into a structured format to be analyzed effectively. Additionally, in this project we checked for duplicate records, combined the datasets into a single dataset and normalized our data, to remove any scaling issues which could affect our model. Furthermore "one-hot" encoding was performed to transform categorical data into numerical format in order to improve model performance.

GitHub URL: labs-jupyter-spacex-Data wrangling.ipynb

EDA with Data Visualization

In this project Exploratory Data Analysis (EDA) was performed in order to discover patterns, find relationships between variables and spot any anomalies that may be present in the dataset.

For this the following methods were used:

1. Bar Chart:

Bar charts has been employed to compare various variables such as launch outcomes and launch sites to highlight whether we could observe any patterns or trends within the data

2. Scatterplot:

Scatterplot was utilised to explore a relationship between two numerical variables such as payload mass and launch success, in order to explore whether there's a correlation between variables and to visualize the impact of one variable on another.

3. Line chart:

Line charts were used in this project to track trends over time such as success rate of Falcon 9 launches across different time periods, so that we could understand the performance trends over specific time periods.

4. Heatmaps:

Heatmaps were used in order to visualize a correlation matrice between multiple variables, in order to help identify strong correlation between variables and use our findings in feature selection.

GitHub URL: edadataviz.ipynb

EDA with SQL

The following SQL queries were utilised within this project:

Aggregate Queries:

- Calculated total number of launches
- Counted successful and failed launches
- Calculated the success rates by launch site and rocket type

Sorting Queries:

- Sorted data to identify trends and outliers
- Ordered launches by date and success rate

Join Queries:

- Joined tables to combine launch records with additional data

Filtering Queries:

- Filtered data to focus on specific launch outcomes

Subqueries:

- Nested queries to calculate average of the data

GitHub URL: jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

Map Objects Created with Folium:

Markers:

- Created markers to represent specific launch sites on the map.
- In order to illustrate the exact launch sites to help users identify where launches had occurured on the map.

Circles:

- Created circles around launch sites to visualize the area around the launch site that might influence launch outcomes.
- In order to illustrate the potential impact of launch sites and their sorrounding area on launch outcomes.

Lines:

- Created lines between launch sites to provide a potential relationship between launch sites.
- In order to provide a visual representation of spatial relationships and dependencies.

GitHub URL: lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

The following were added to our interactive Dashboard:

Plots and Graphs:

Pie Chart:

- Displays the proportion of successful and failed launches in order to visualize the success rate and their launch sites
- Provides an easy and intuitive way to present our findings to stakeholders.

Scatter Plot:

- Illustrates the relationship between payload mass and launch sites and gives the user an ability to explore how payload mass influences mission outcomes

Launch Site Dropdown:

- Implemented a dropdown feature to allow users to select specific launch sites in order to help with filtering and analysis of data based on geographical locations. This allows for thorough analysis of different launch sites

Range Slider:

- Slider allows the user to explore how payload mass impacts the launch success.

GitHub URL: spacex-dash-app.py

Predictive Analysis (Classification)

Building our model

1. Data Preprocessing:

- Standardized features to ensure all variables contribute equally, then the data was split into training and test sets for model validation.

2. Model Selection:

- Explored multiple classification algorithms such as: SVM, Decision Trees and K-Nearest Neighbors. The models were selected based on the project requirements.

3. Hyperparameter Tuning:

- Performed GridSearchCV to find the optimal hyperparameters for the models, with which we tuned our models, such as C (SVM) , max_depth (Decision Trees) and n_neighbors (KNN)

4. Model Evaluation:

- Evaluated our models using cross-validation techniques to ensure robustness and generalizability. Additionally, we utilised metrics such as accuracy, precision, recall and F1 score to examine our model performance

5. Model Improvement:

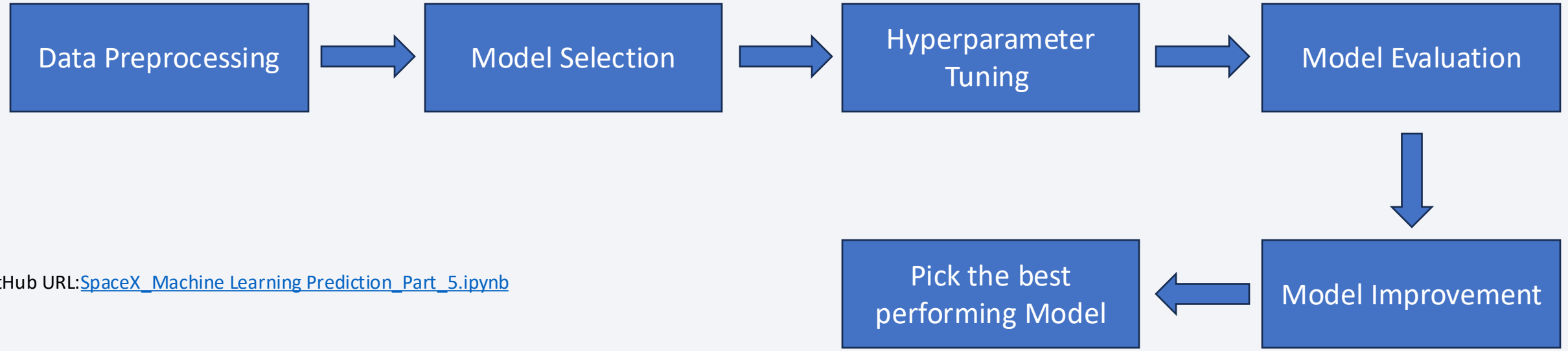
- Adjusted our models based on insights from validation results. Further fine tuned our hyperparameters to improve our models predictive accuracy and reliability.

Predictive Analysis (Classification) - Continued

Building our model

6. Best Performing Model:

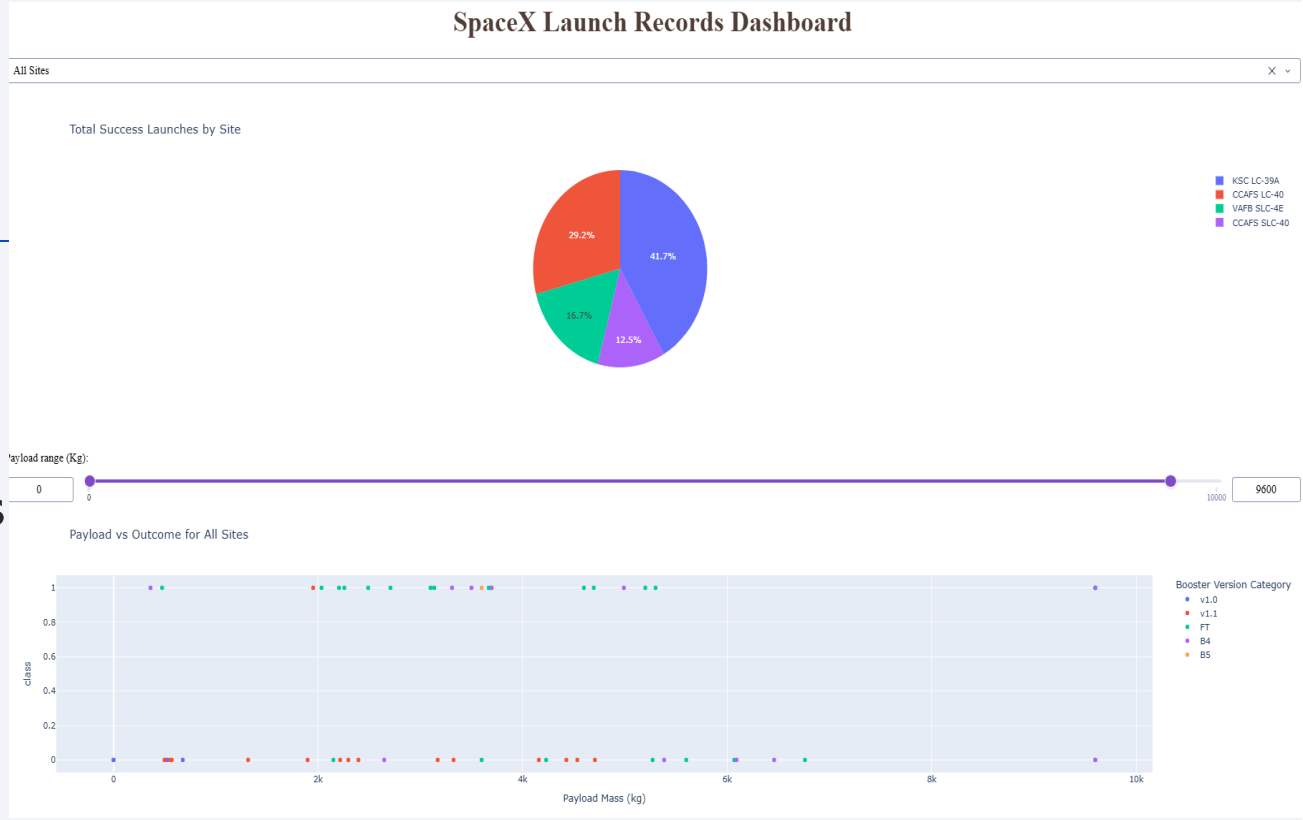
- Identify and choose the model that has the highest accuracy on the test set. Evaluate the training and test set performance to avoid overfitting and to enable real world performance.

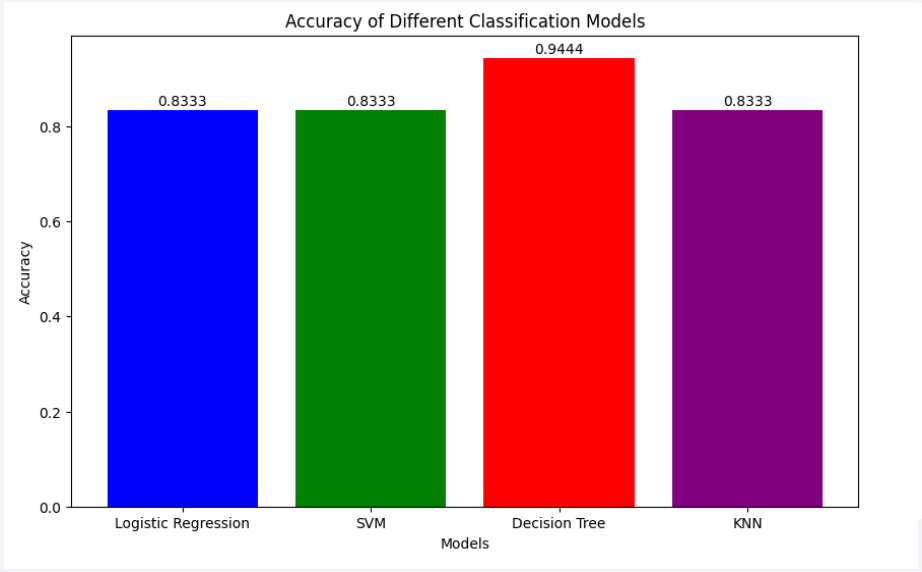


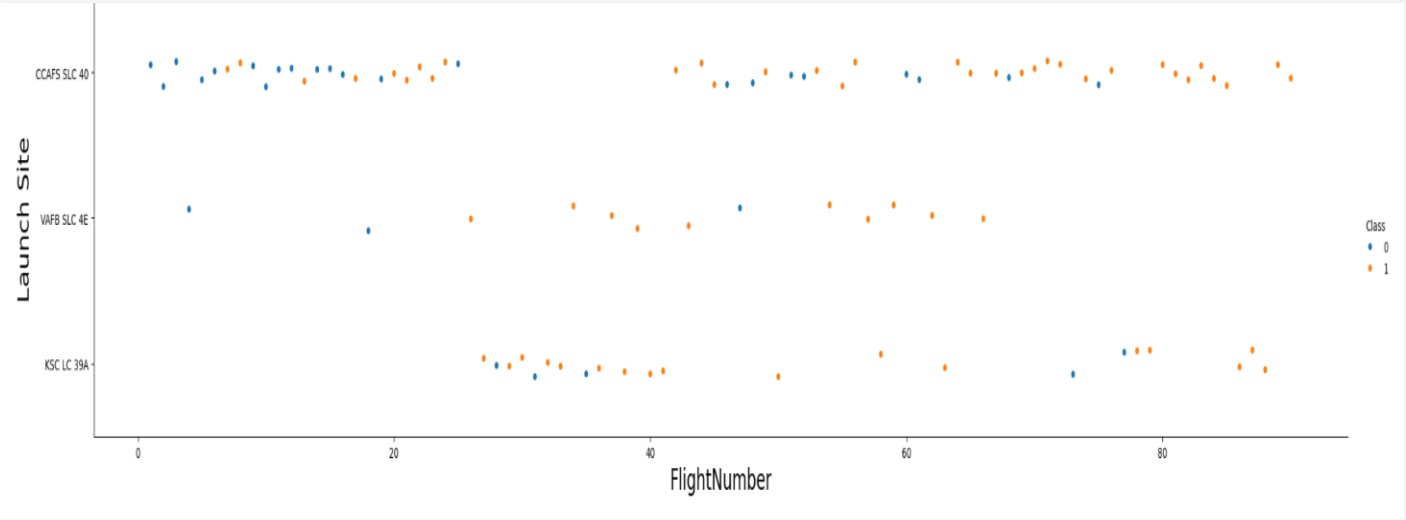
GitHub URL:SpaceX_Machine Learning Prediction_Part_5.ipynb

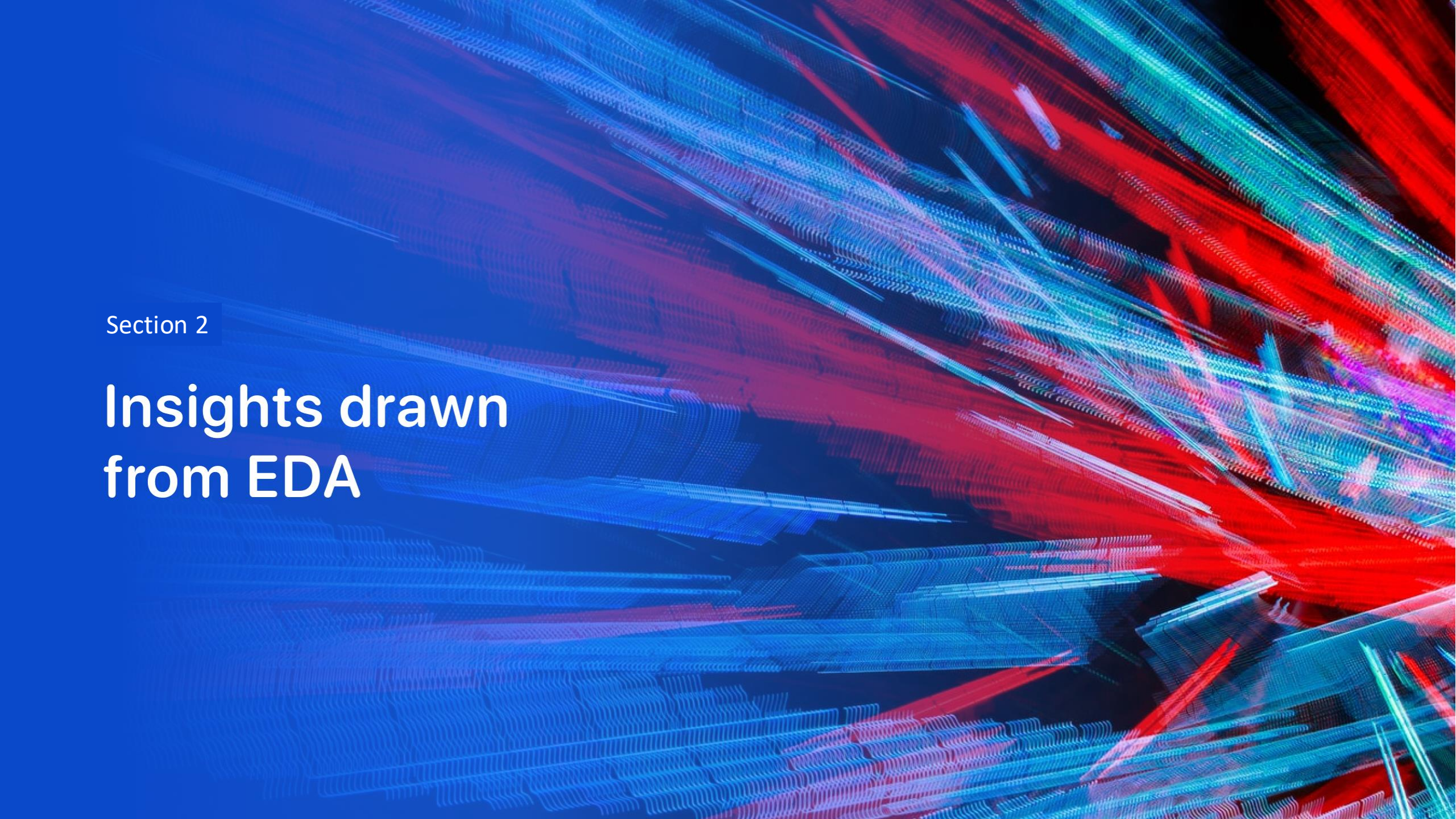
Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results



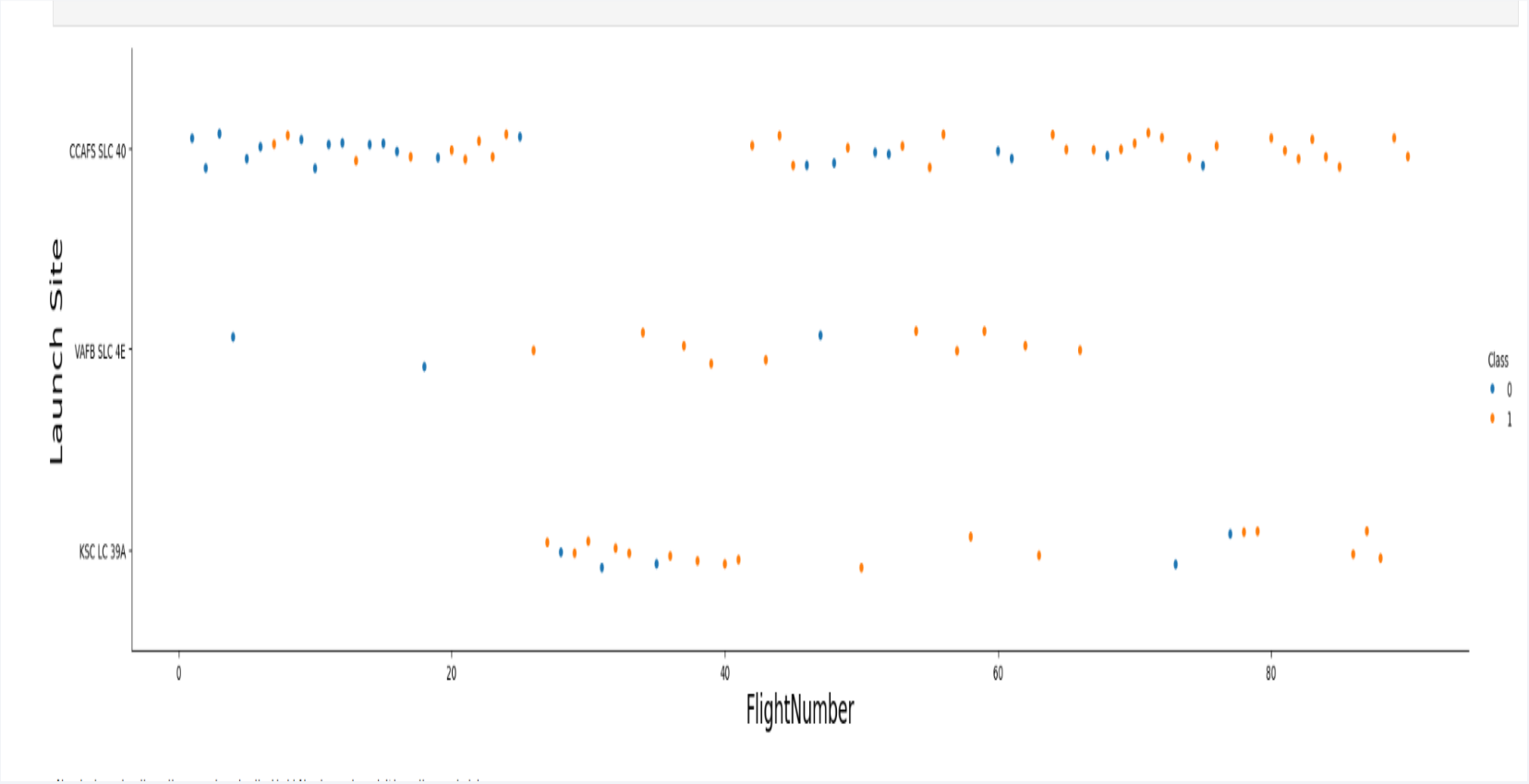





Section 2
Insights drawn from EDA

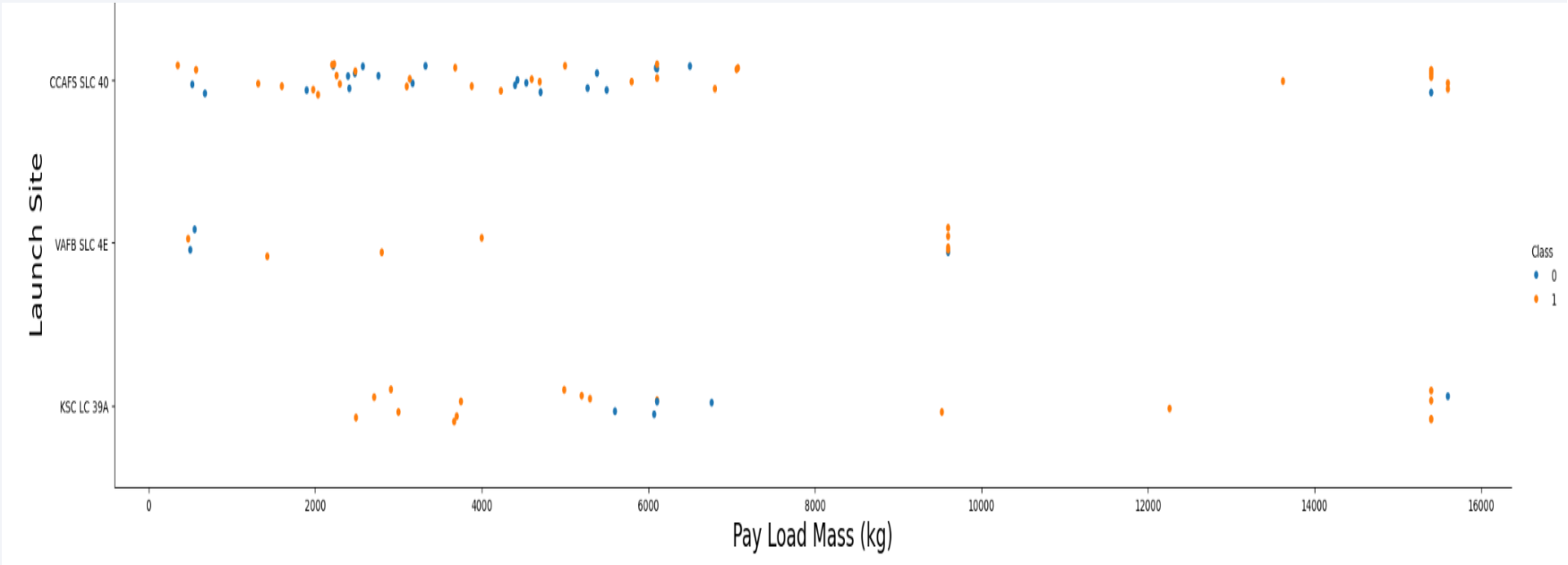
Flight Number vs. Launch Site

- Our findings demonstrate that there is a mix of successful and unsuccessful landings at the launch sites which indicate that other variables may influence the landing outcomes.
- Additionally, there's a wide range of flight numbers at our launch sites, with a trend of more successful landings at the higher end, demonstrating that perhaps launch sites don't have a linear correlation with success rate.



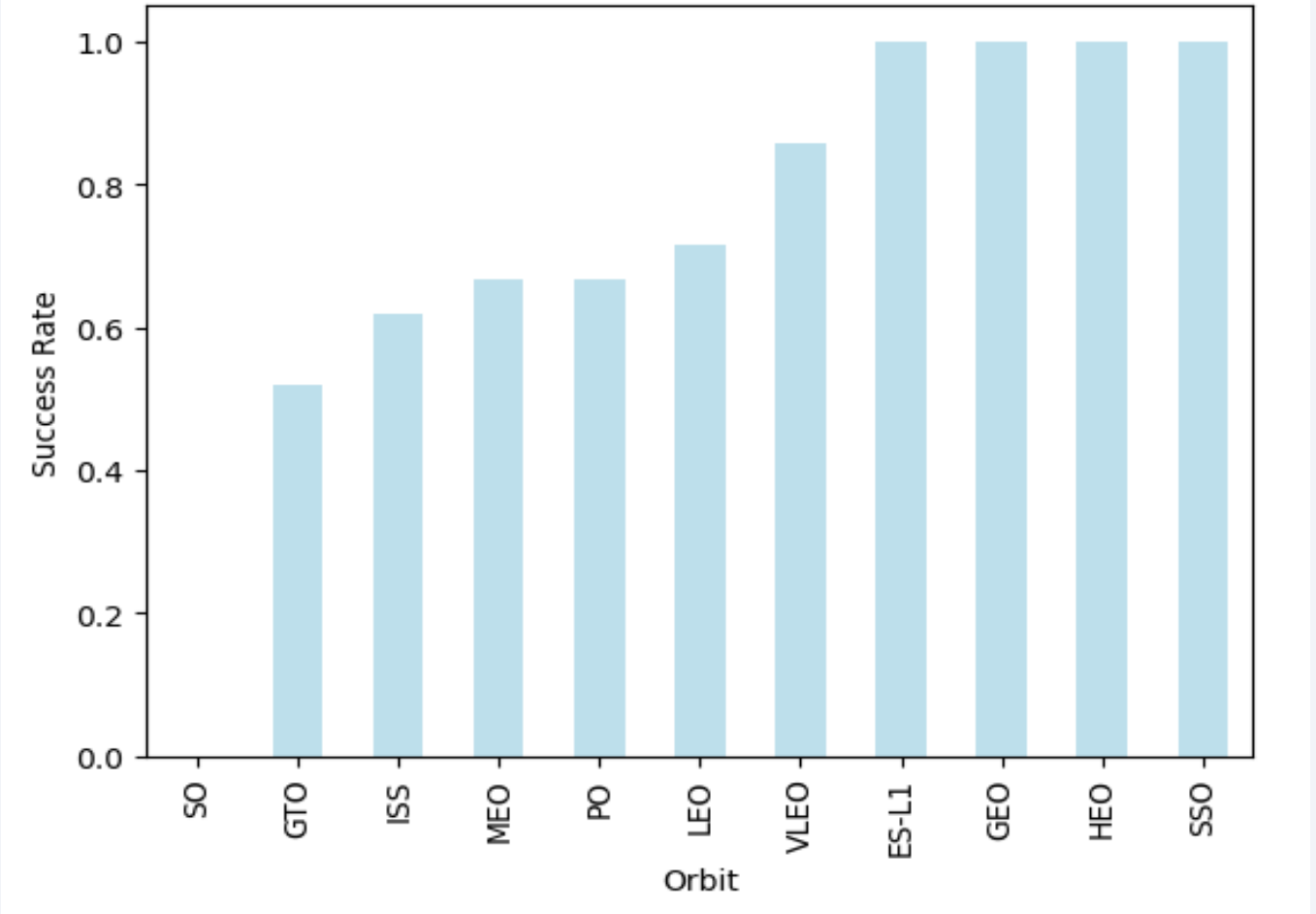
Payload vs. Launch Site

- Our findings suggest that different launch sites were picked depending on the payload, such as CCAFS SLC 40 being utilised for payloads with a mass below 10,000kg, whilst the other launch sites were used for heavier payloads. Additionally, KSC LC 39A and VAFB SLC 4E were utilised for launches with heavier payloads.



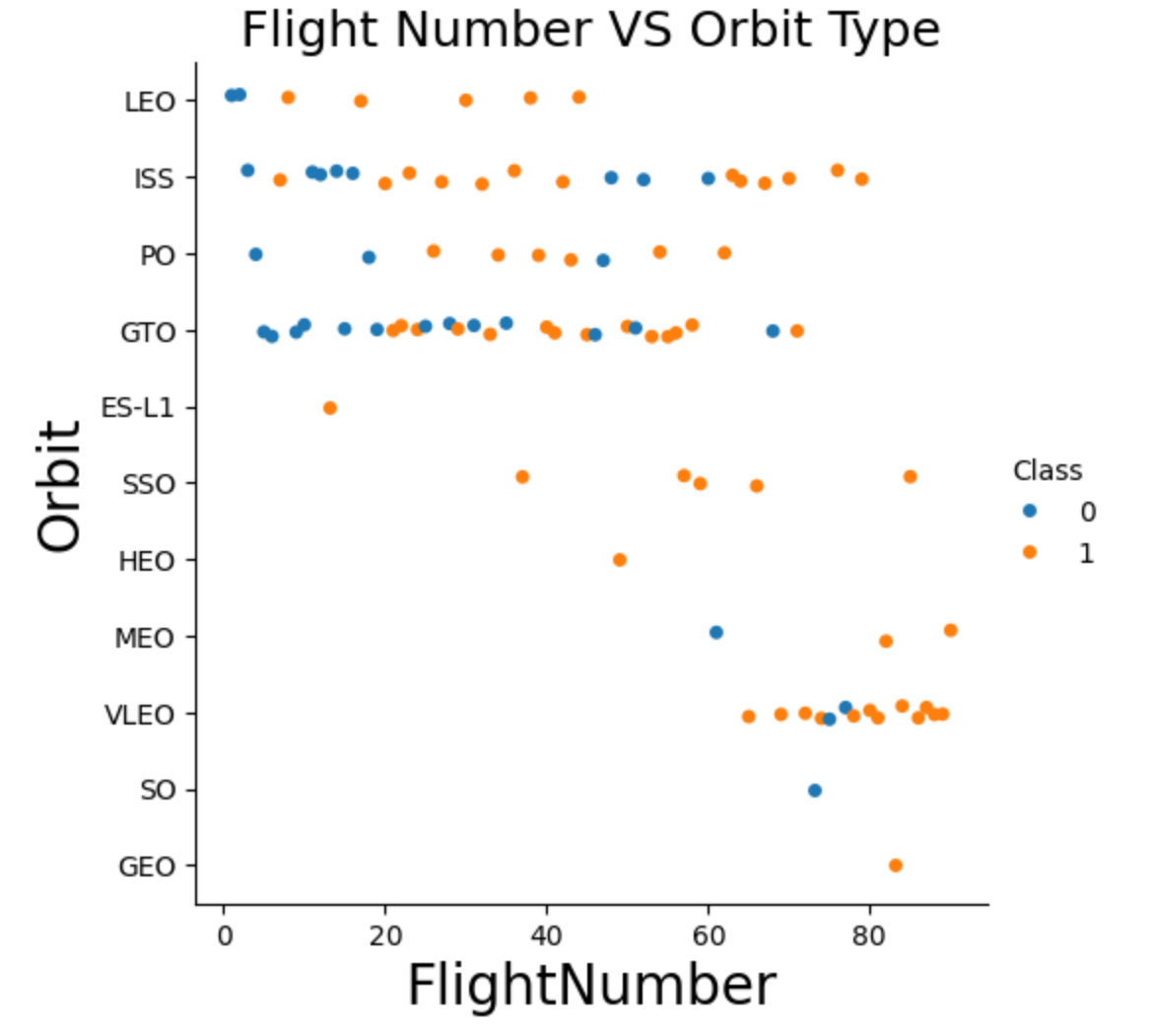
Success Rate vs. Orbit Type

Our findings suggest that orbits of ES-L1 to SSO are the right type for successful first stage landings due to high success rate as demonstrated by the graph. However, orbit GTO suggests that it may not be a suitable type for first stage landings due to a poor success rate, as demonstrated by the graph



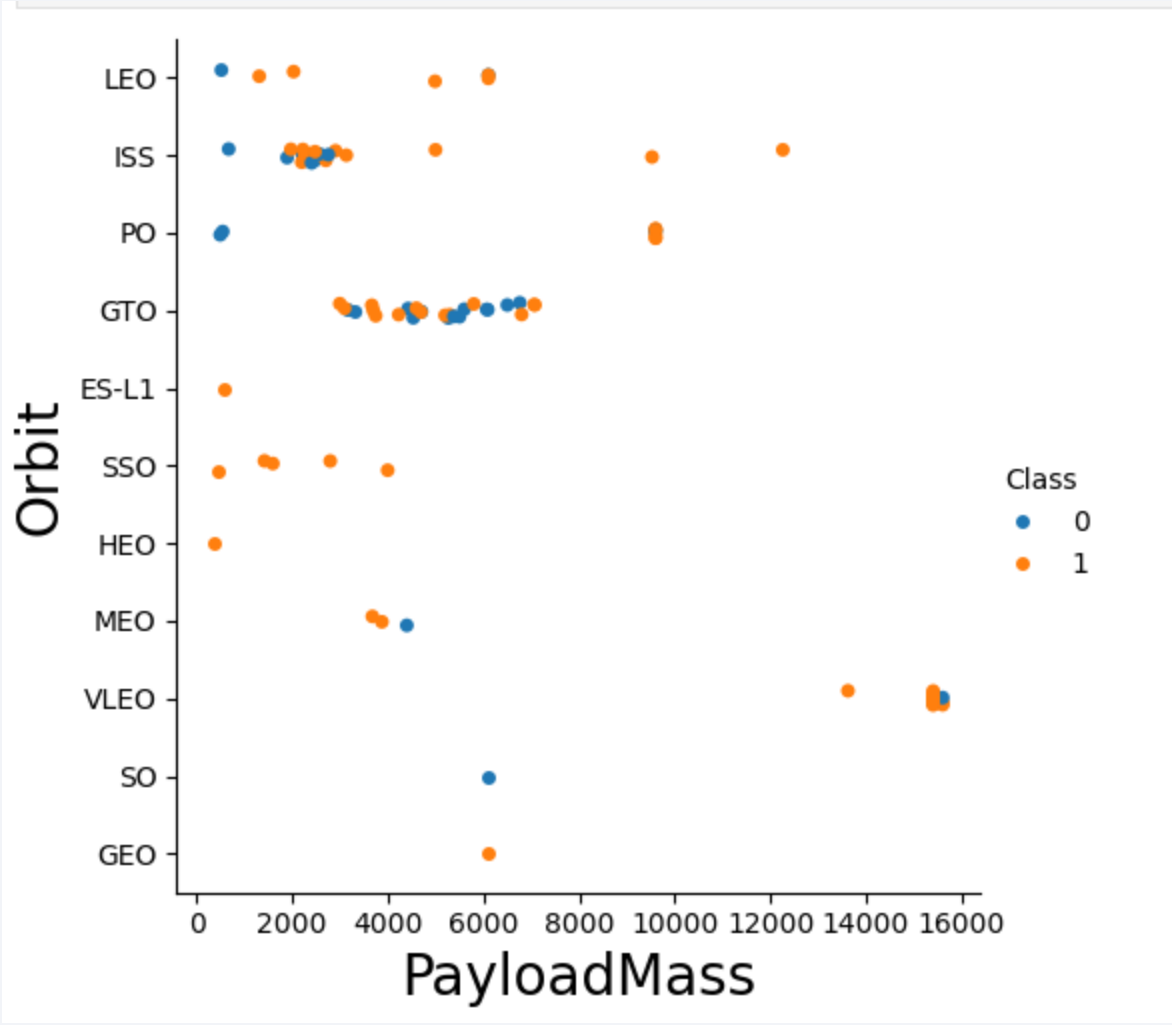
Flight Number vs. Orbit Type

- Our findings suggest that the success rate of the rocket improves significantly with more flight numbers, which may demonstrate that increased experience from higher number flights lead to higher success rate.



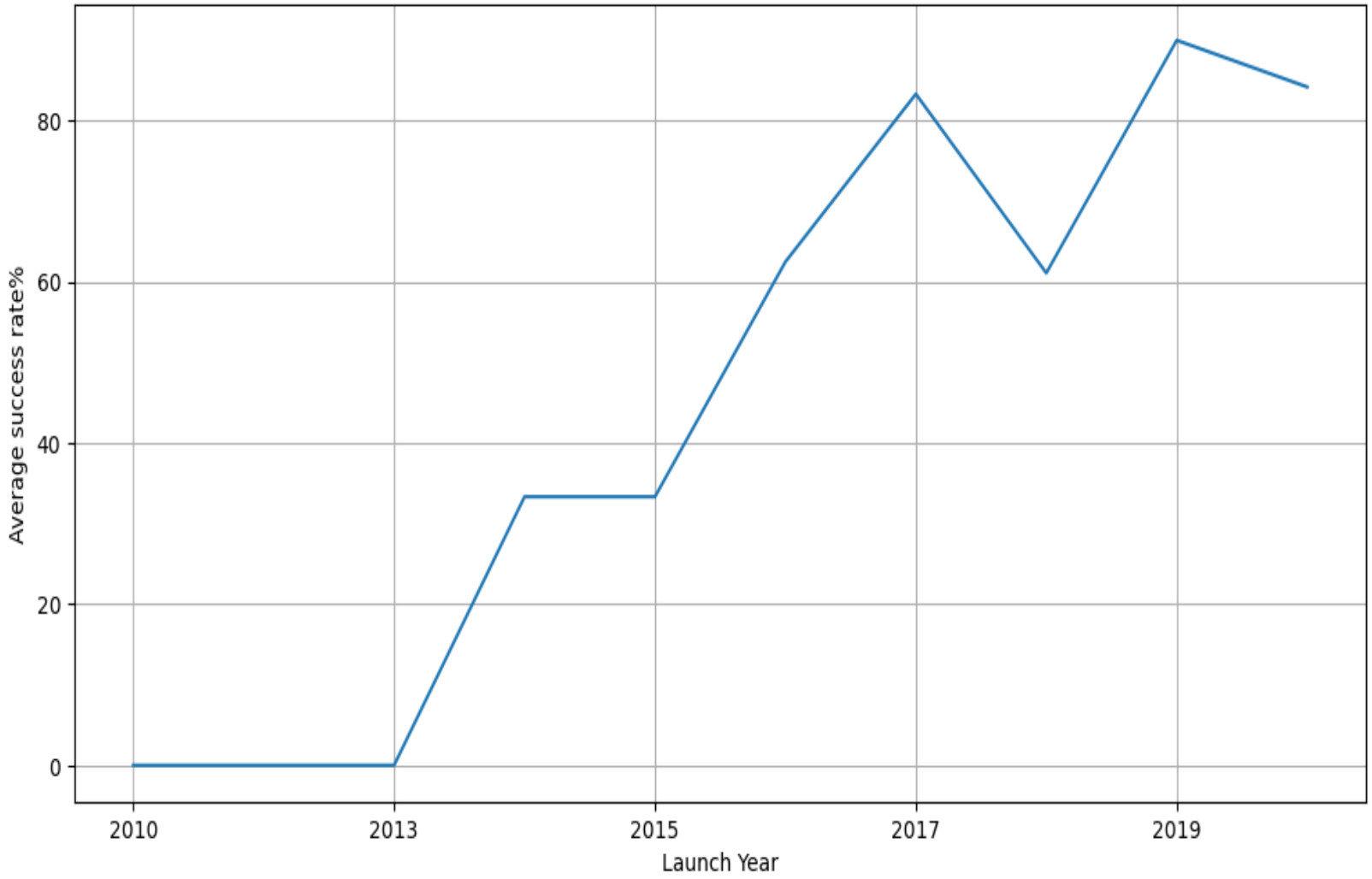
Payload vs. Orbit Type

- The graph demonstrates a lack of correlation due to the spread of outcomes, which may suggest that orbit type and payload mass may not directly influence launch outcomes.



Launch Success Yearly Trend

- The graph shows us a significant increase in success rate over the time period, which demonstrates us increasing success in Falcon 9 launches, especially if we compare success rate at 2013 to 2020, with an improvement of over 80%



All Launch Site Names

In [11]:
```
%%sql
select Distinct Launch_site from SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Out[11]:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Display 5 records where launch sites begin with the string 'CCA'

[12]:
```
%%sql
select * from SPACEXTABLE where Launch_site like "CCA%" limit 5;
```

```
 * sqlite:///my_data1.db
Done.
```

[12]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%%sql
select SUM(PAYLOAD_MASS__KG_) from SPACEXTABLE where customer == 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

SUM(PAYLOAD_MASS__KG_)
45596

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
]:
%%sql
select AVG(PAYLOAD_MASS__KG_) from SPACEXTABLE where Booster_Version == 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

]:

AVG(PAYLOAD_MASS__KG_)
2928.4

First Successful Ground Landing Date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint:Use min function

15]:
```
%%sql
select Date from SPACEXTABLE where Landing_Outcome == 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
Done.
```

15]:

Date
2015-12-22
2016-07-18
2017-02-19
2017-05-01
2017-06-03
2017-08-14
2017-09-07
2017-12-15

Successful Drone Ship Landing with Payload between 4000 and 6000

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%%sql
select Booster_Version from SPACEXTABLE where (Landing_Outcome == 'Success (ground pad)') and (PAYLOAD_MASS__KG_ between 4000 and 6000);
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1032.1
F9 B4 B1040.1
F9 B4 B1043.1

Total Number of Successful and Failure Mission Outcomes

List the total number of successful and failure mission outcomes

```
%%sql
select count(Mission_Outcome) AS Total from SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Total
101

Boosters Carried Maximum Payload

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
]:
%%sql
select Booster_Version from SPACEXTABLE where PAYLOAD_MASS__KG_ == (select MAX(PAYLOAD_MASS__KG_) from SPACEXTABLE);
```

```
 * sqlite:///my_data1.db
Done.
```

]:

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
%%sql
select substr(Date,6,2) as Months_Name, Booster_Version, Launch_Site, Landing_Outcome, Date from SPACEXTABLE where substr(Date,0,5) =='2015' and (Landing_Outcome == "Failure (drone ship)");
```

```
 * sqlite:///my_data1.db
Done.
```

Months_Name	Booster_Version	Launch_Site	Landing_Outcome	Date
01	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)	2015-01-10
04	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)	2015-04-14

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

In [21]:
```
%%sql
select Landing_Outcome, Count(*) AS "Count" From SPACEXTABLE WHERE Date between '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY COUNT(*) DESC;
```

```
* sqlite:///my_data1.db
Done.
```

Out[21]:

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1


Section 3
Launch Sites
Proximities Analysis

Marking all launch sites on a map

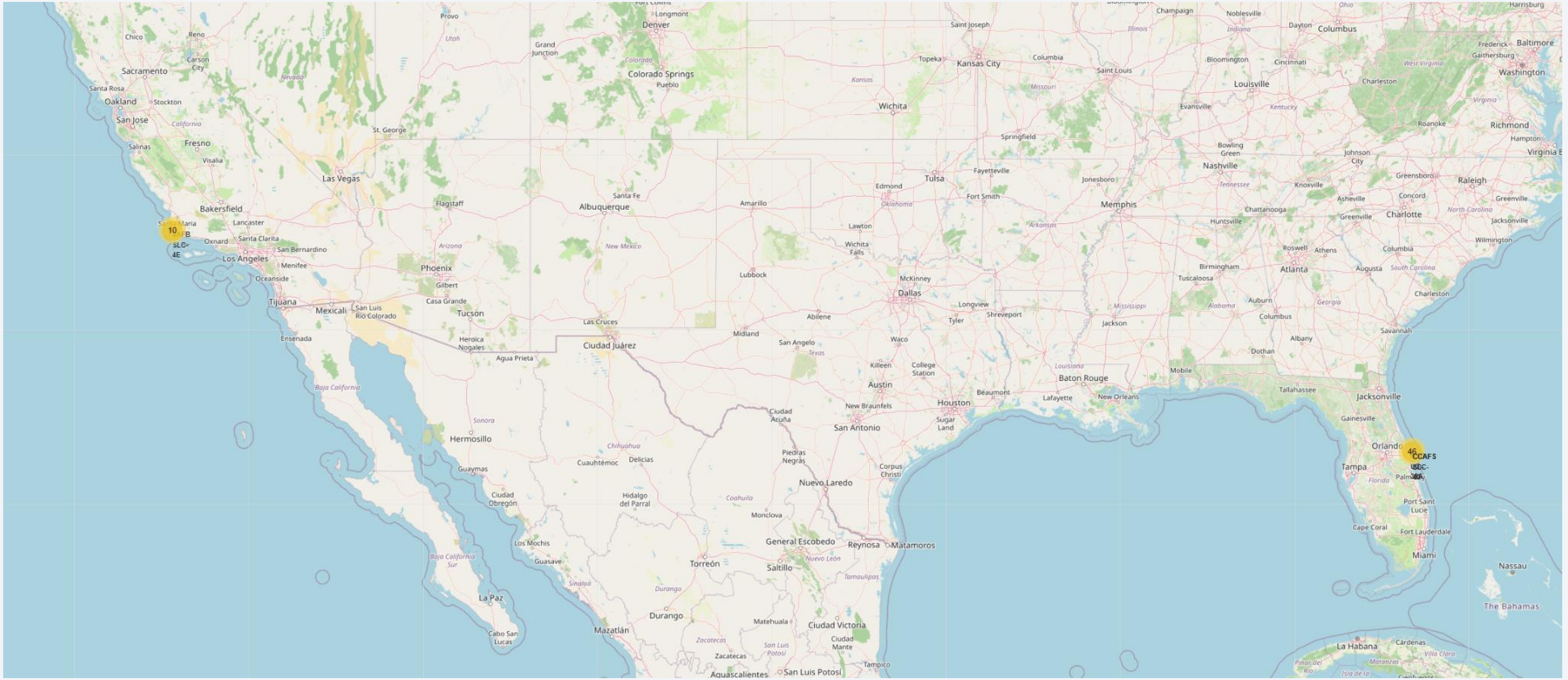
The following questions were utilized to gather insights into launch sites:

1. Are all launch sites in proximity to the Equator line?
2. Are all launch sites in very close proximity to the coast?

Our findings demonstrate that not all launch sites are in close proximity to the Equator, such as the launch site at Vandenberg Air Force Base (VAFB SLC-4E), which is located further from the Equator compared to the other sites in Florida.

Additionally, our findings show us that the launch sites are in close proximity to the coast, such as the Vandenberg Air Force Base (VAFB SLC-4E) which is located near the coast of California.

Marking all launch sites on a map



Mark the success/failed launches for each site on the map

To make analysis of the launch data more intuitive we added additional markers, which display the success rate of each launch site. This measure allows the user to examine any potential patterns with ease. Furthermore, the colour and the popup information from the markers aid the user in analysing the SpaceX launch data.

For example, launch site CCAFS LC-40 has multiple red and green markers (red for failed and green for success), which make it easier to quickly digest the success rate of that launch site. We employed this technique as it allows for a quick visual feedback on the performance of each launch site.

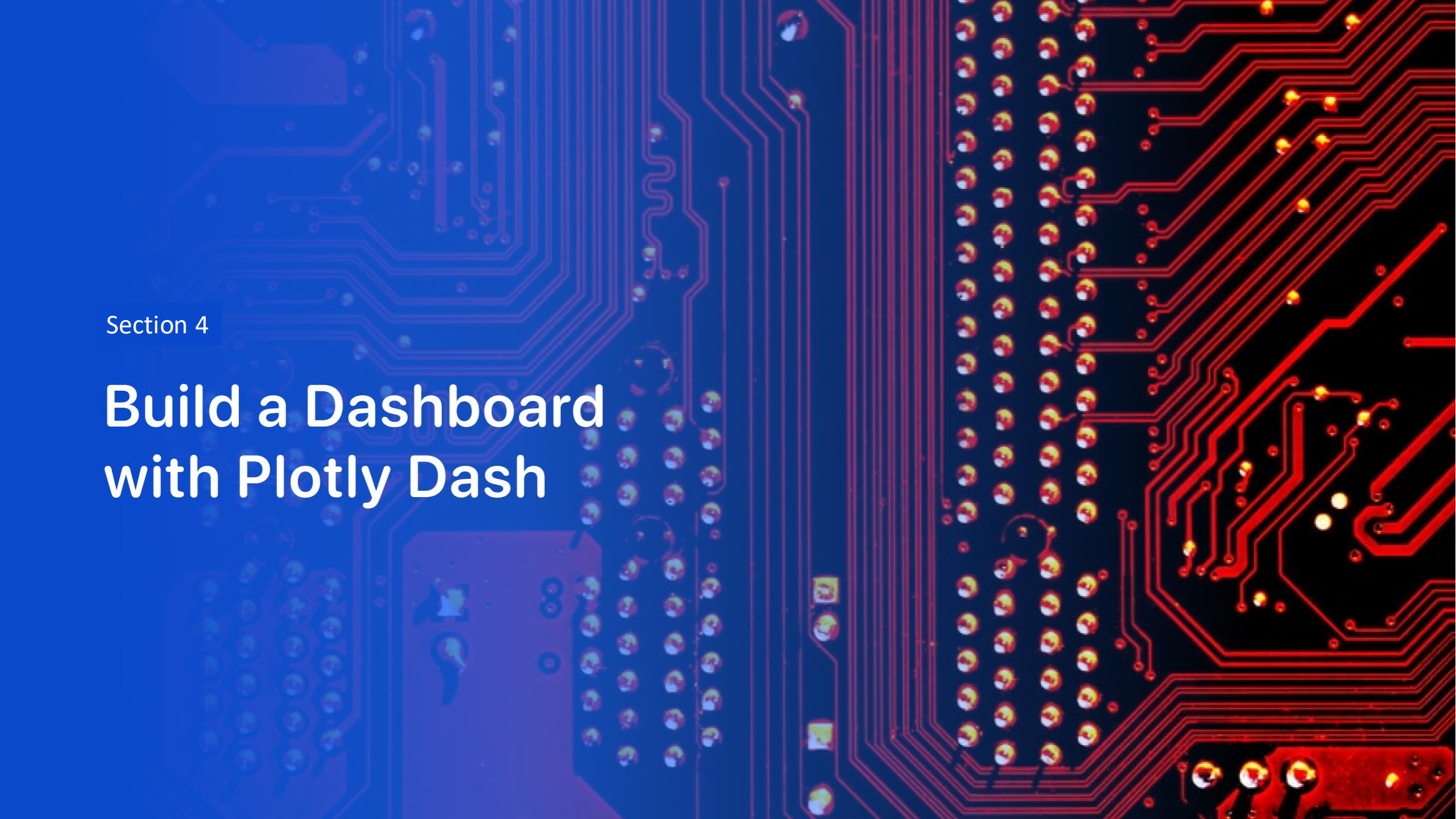
Mark the success/failed launches for each site on the map



Calculate the distance between a launch site to its proximities

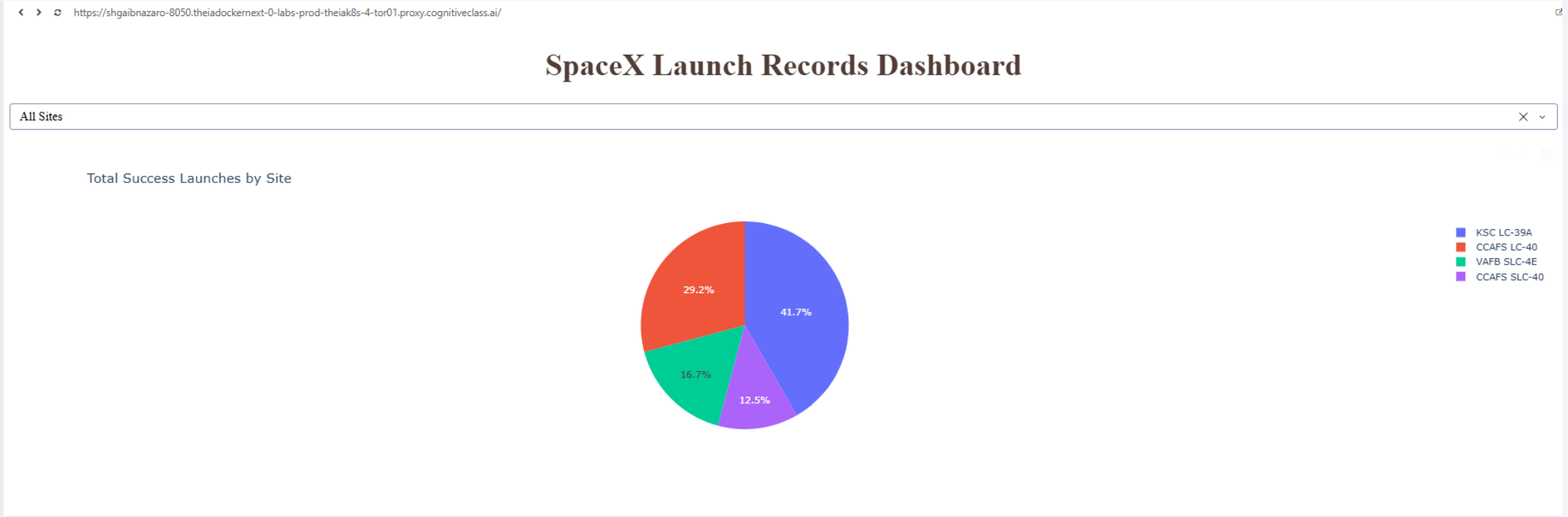
- We employed the following technique displayed in the graphs to clearly mark the proximity of launch sites to it's proximities, such as the coast or the closest city. This was done to gather geographical information on launch sites and their surrounding area in order to accurately gauge which factors influence the launch decisions.




Section 4
Build a Dashboard with Plotly Dash

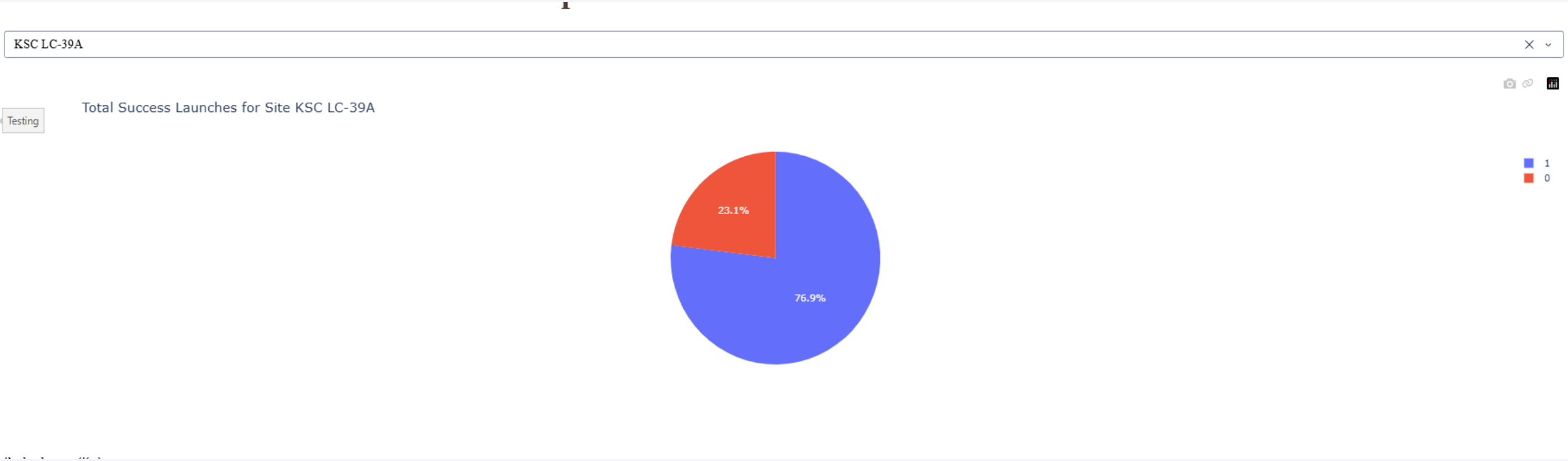
Ratio of all successful launches per sites

- Our findings indicate that KSC LS-39A launch site has the highest number of successful launches, consisting of 41.7% (as demonstrated by the graph) of all successful launches. We can infer that KSC LS-39A will best choice for potential launches.



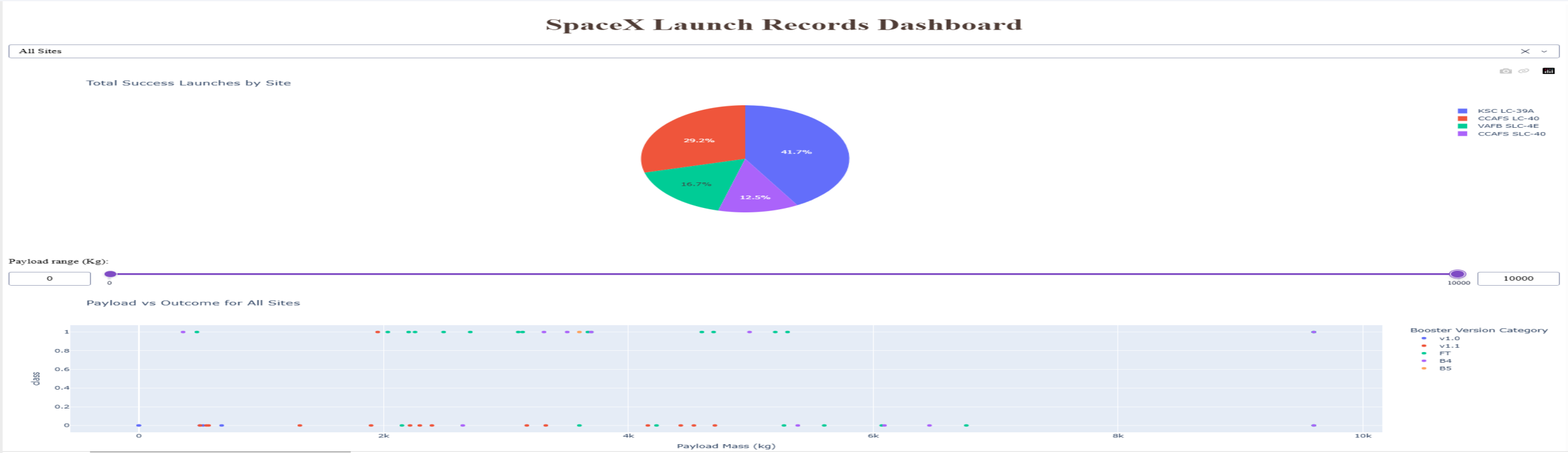
Ratio of successful launches per site

- Our finding suggest that KSC LC-39A has the highest ratio of successful to failed launches. This is demonstrated by the graph were the successful launches made up 76.9% whilst failed launches made up 23.1%. From this we can infer that KSC LC-39A is the best launch site due to its high success rate.



Effects of payload and success for all sites

- Our findings from the graph below suggest that the booster version "FT" has the highest success rate, as it contains the highest number of points across the success class (class = 1), whilst the booster version "v1.0" has the lowest success rate. However, the spread of points over success rate and lack of clear trend indicate that there may be no or very weak correlation between payload mass and success rates.



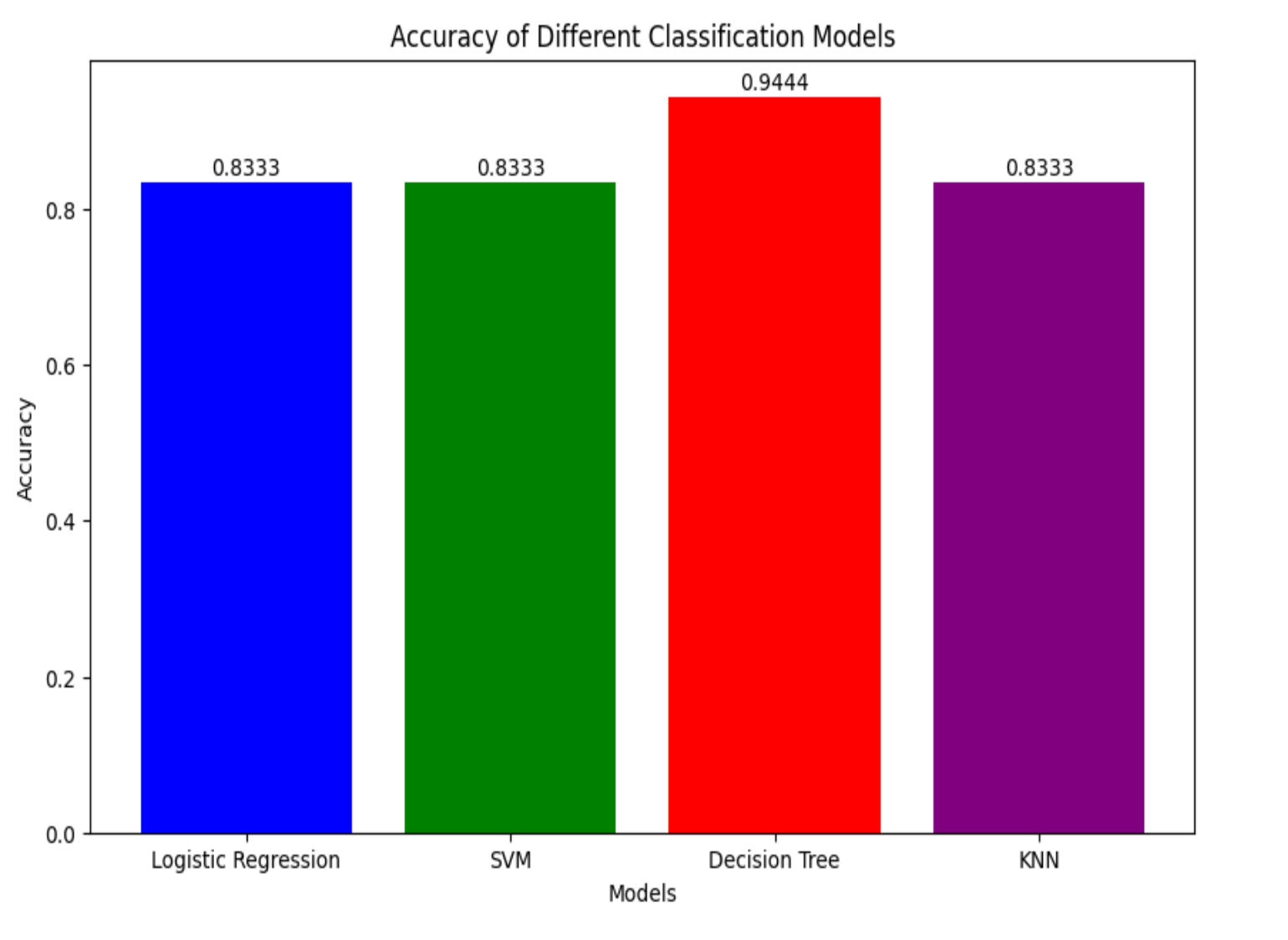
Section 5

Predictive Analysis (Classification)



Classification Accuracy

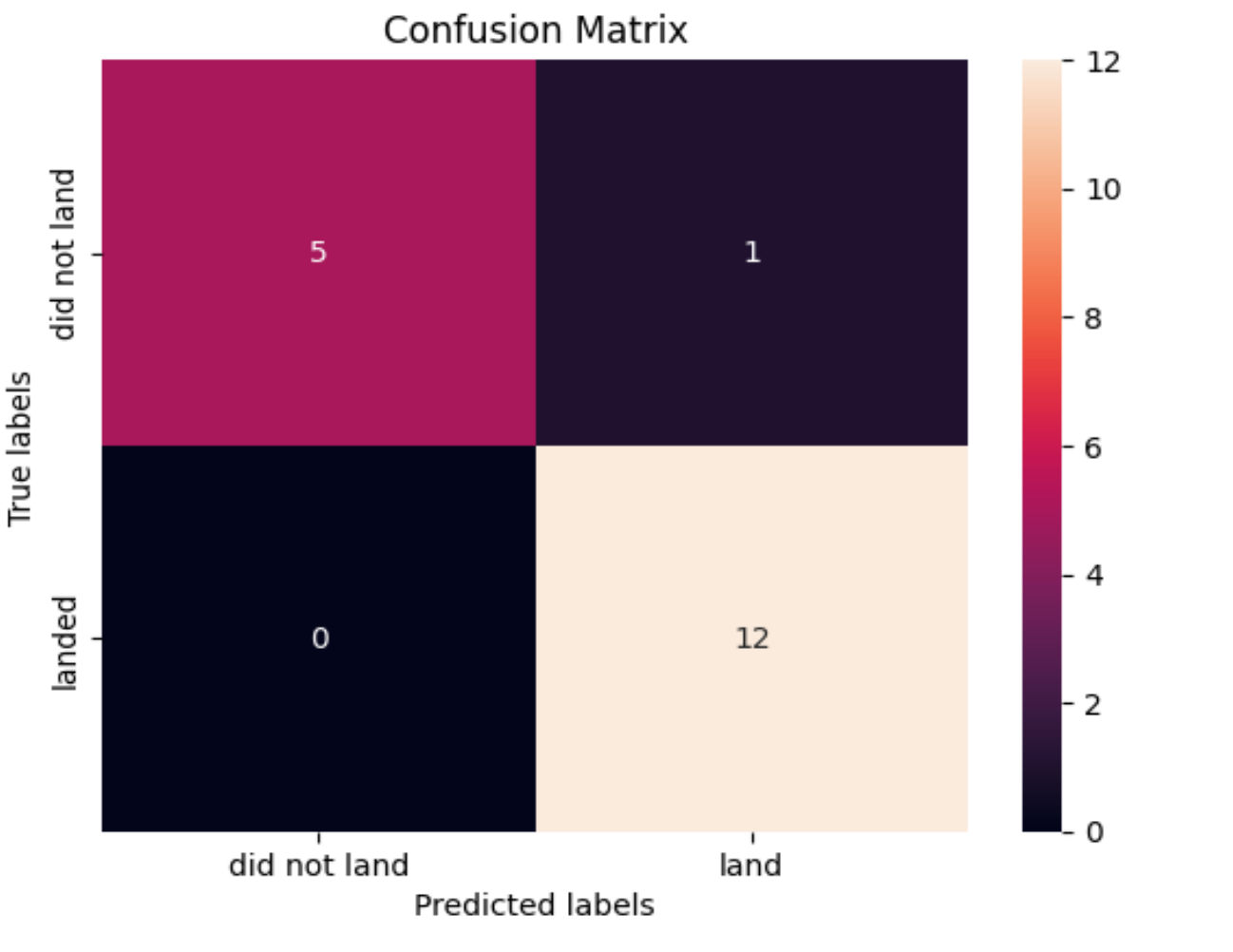
Based on our findings, the best classification model is the Decision Tree model as it has the highest classification accuracy (of 0.944). This indicates that Decision Tree model is the best model for this dataset compared to other classification models.



Confusion Matrix

- We believe this model to be a good fit based on the confusion matrix, which shows us a high accuracy score of 94.44% with a significant number of true positives and true negatives, indicating its effectiveness in predicting Falcon 9 first stage landings. Additionally, the lack of false negatives, indicate the models reliable predictions of successful landings. This is pivotal for this project as ensures no wasted launches and keeps the launch costs low. Furthermore, the low false positive score, indicate this model to suitable for real world application. Finally, this model demonstrates a good fit for predicting real world launches, which is crucial in ensuring low costs in a big against SpaceX by accurately predicting successful landings and not wasting any potential resources.

Confusion Matrix



Conclusions

Point 1: From our research we can identify that KSC LC-39A has the highest success rate compated to other sites within this project, making up 41.7% of all successful launches.

Point 2: From our research we can identify booster version "FT" as the most suitable booster version due to high success rate across different payload masses.

Point 3: From our research we can identify no clear pattern between payload mass and success rates, which suggests that other factors may influence launch outcomes such as launch site or booster version.

Point 4: We employed interactive data visualization tools using "Folium" and "Plotly Dash" to enable easy understanding of data for potential stakeholders in order to make informed decisions based on our findings.

In conclusion, our research via predictive analysis and interactive visualization have highlighted key factors that influence SpaceX's successful launches and provided clear guidelines in order to improve potential launches in the future. We believe this data can help to reduce costs for future launches and help to compete against SpaceX in making reusable rocket technology.

Thank you!

